MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-BVOC (HONOURS) Regular EXAMINATION

THIRD SEMESTER

MG3MDCREM200- IKS for Renewable Energy Transformation in Kerala

(2025 ADMISSION ONWARDS)

Duration:1.5 hr **Maximum Marks:50**

Rememder(K) , *Understand(U), Apply(A), Analyse(An),* Evaluate(E), Create(C), Skill(S), *Interest(I) and Appreciation(Ap)*

PART A

Short answer

Answer any 30 questions

Each question carries 1 marks

1. Identify the knowledge system that represents the traditional knowledge of India. [U, CO1]

2. Mention any one significant contribution of Kerala to IKS. [R, CO2]

3. Name the traditional system of medicine associated with Kerala and South India. [R, CO1]

4. Give one characteristic feature of Ayurveda. [U, CO1]

5. Cite one example of a traditional water management practice in Kerala. [R, CO1]

6. Identify the role of temples in traditional astronomical practices. [U, CO1]

7. Write the meaning of the term Panchangam. [U, CO1]

8. List any two components of Panchangam. [R, CO1]

9. Indicate one application of traditional calendars in agriculture. [U, CO2]

10. Differentiate between solar and lunar calendars. [An, CO1]

11. Mention one energy-related application of traditional astronomical knowledge. [U, CO2]

12. Point out one way in which IKS can contribute to modern renewable energy policies. [A, CO4]

13. Identify the ancient Indian practice associated with salutations to the Sun. [U, CO1]

14. Name one ancient Indian astronomical system used for understanding solar movements. [R, CO1]

15. Mention one important solar temple in India. [R, CO2]

16. Locate the famous Konark Sun Temple. [R, CO1]

17. Identify the basic principle involved in traditional sun-tracking practices. [U, CO2]

18. Give one example of an ancient structure associated with solar observation. [R, CO1]

19. Mention one possible connection between traditional solar knowledge and modern solar PV systems. [U, CO3]

20. Specify one application of solar energy in traditional practices. [U, CO2]

21. Identify the major purpose of integrating ancient solar knowledge with modern solar technology. [U, CO3]

22. Name the traditional Kerala architectural system known for its central courtyard. [R, CO2]

23. Mention one feature of a Nalukettu that promotes natural ventilation. [R, CO2]

24. Identify the passive solar feature commonly associated with traditional Kerala architecture. [U, CO2]

25. Give one advantage of passive solar lighting. [U, CO3]

26. Name any two biomass resources commonly available in Kerala. [R, CO2]

27. Mention one traditional use of cow dung as an energy resource. [R, CO2]

28. Identify the process through which organic matter is converted into biogas. [U, CO3]

29. Give one example of agricultural waste that can be used for energy generation. [R, CO2]

30. Mention one advantage of community-scale biomass energy systems. [U, CO3]

31. Identify one difference between traditional biogas systems and modern biogas plants. [An, CO3]

32. Write the full form of KSEB. [R, CO3]

PART B

Short Essay

Answer any 4 questions

Each question carries 5 marks

33. Trace the historical evolution of Indian Knowledge Systems and examine Kerala's contribution to its development. [An, CO1]

34. Compare Siddha and Ayurveda with respect to their basic principles and traditional significance. [An, CO1]

35. Describe the structure of Panchangam and examine the relevance of traditional solar-lunar calculations in agriculture and energy-related activities. [An, CO1]

36. Discuss the traditional Indian knowledge of solar astronomy with reference to Surya Namaskara, sun-tracking practices and solar temples. [U, CO2]

37. Analyse the architectural principles of Nalukettu and explain how they facilitate passive solar lighting and natural ventilation. [An, CO2]

38. Discuss the role of biomass and bioenergy in sustainable agriculture. Explain the utilisation of cow dung, coconut shells and agricultural residues for energy generation. [U, CO3]